

# Off-Grid Spectrum Cartography via Deep Generative Models

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**Abstract**—This work studies radio map estimation (also known as spectrum cartography) from off-grid measurements over a continuous spatial domain. Classical approaches to off-grid radio map sensing rely on kernel or polynomial-based interpolation methods. More recently, learned deep generative models (DGMs) have demonstrated strong empirical performance in recovering complex radio maps from highly undersampled measurements, owing to their expressive power. However, DGMs are typically trained on gridded, pixel-based data, and how to coherently integrate them with off-grid sensing remains unclear. We employ a multi-dimensional *discrete cosine transform* (DCT) representation for modeling radio maps over continuous space. DCT provides a computationally efficient and theoretically grounded basis that bridges off-grid measurements with pre-trained DGMs of the radio map, leading to a simple, compact formulation that can be tackled by off-the-shelf optimization algorithms. Numerical simulations demonstrate the effectiveness of the proposed approach in off-grid radio map estimation scenarios.

**Index Terms**—Spectrum cartography, radio map estimation, off-grid sensing, deep generative models

## I. INTRODUCTION

Radio map estimation, also known as *spectrum cartography* (SC), aims to recover the spatial–frequency variation of received radio power over a region of interest. Radio maps provide rich radio frequency (RF) information across space and frequency, assisting various applications in wireless resource allocation and networking [1]–[3]. SC has attracted growing attention in the sensing and communications communities; see, e.g., [4]–[13]. Early approaches primarily relied on kernel- and polynomial-based interpolation techniques [7], [14]. More recently, matrix and tensor completion methods have been employed for SC [9], [10], [15], yielding improved performance and theoretical recoverability guarantees. In parallel, deep generative models (DGMs) have been integrated into SC frameworks [5], [8], [16], demonstrating further gains in complex and challenging propagation environments, such as scenarios with heavy shadowing.

An important characteristic of radio maps is that the underlying signals are defined over a continuous spatial domain. Consequently, measurements are acquired at arbitrary spatial locations rather than on discretized, index-based grids arising from matrix or tensor representations. Early kernel- and

polynomial-based interpolation approaches (e.g., [4], [7], [14]) naturally accommodate off-grid sensing. Moreover, matrix and tensor completion methods can be flexibly combined with interpolation mechanisms—such as interpreting on-grid (one-shot or iterative) interpolations through low-rank models—to handle this setting; see, e.g., [9], [10], [17], [18]. Nonetheless, interpolation methods are good at gathering information when the RF environment is relatively smooth (e.g., in outdoor or free space), yet often struggle to perform well when the RF environment characteristics change drastically across the field of interest (e.g., in indoor or urban areas). As a consequence, relying on interpolated on-grid estimation to perform matrix/tensor based SC hinges on the accuracy of interpolation, which may not be ideal. Like matrix and tensor-based methods, DGMs are typically learned on discretized, pixel-based representations. As a result, how to effectively extend DGM-based spectrum cartography methods (e.g., [5], [8]) to off-grid sensing scenarios—without resorting to interpolation-based on-grid point estimation—remains an open challenge.

We propose a DGM-based framework for off-grid spectrum cartography that avoids explicitly interpolating off-grid measurements to estimate on-grid values. By modeling radio maps as multi-dimensional signals with bounded spatial support, we adopt a *discrete cosine transform* (DCT)–based high-order tensor representation [19] to bridge grid-trained DGMs with off-grid measurements. Specifically, on-grid radio maps generated by a DGM are mapped to their DCT coefficient representation, which induces a continuous spatial model capable of evaluating the radio map at arbitrary locations. This construction enables a unified SC loss in which DGMs are embedded in the latent domain, allowing existing DGM-based on-grid SC optimization algorithms to be directly repurposed for off-grid sensing. Moreover, since DCT bases provide provable approximation guarantees for signals with bounded support, the proposed approach admits theoretical justification from classical sampling and approximation theory. Numerical simulations demonstrate that the proposed method consistently outperforms existing approaches by nontrivial margins.

**Notation.** We largely follow established conventions in signal processing and the SC literature; see, e.g., [5], [8], [15].

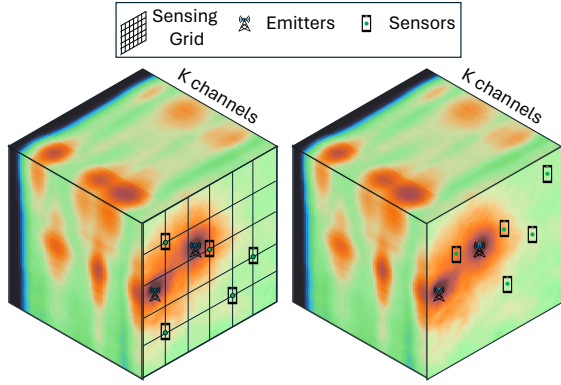


Fig. 1. Illustration of on-grid SC (left) and off-grid SC (right) in spatially continuous radio maps.

## II. BACKGROUND

### A. Problem Setup

We consider a spatially continuous radio map over  $K$  frequency bins. For each frequency  $k$ , the function  $f_k(u, v) : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$  represents the received signal strength at spatial location  $(u, v)$ , where  $u, v \in [0, 1]$ . The spatial domain is normalized to  $[0, 1]^2$  without loss of generality. When there are  $R$  emitters present in the region and when the spectral band of interest is relatively narrow relative to the carrier frequency, the radio map admits a decomposition-based representation (see, e.g., [4], [7], [15])

$$f_k(u, v) = \sum_{r=1}^R s_r(u, v) c_{r,k}, \quad (1)$$

where  $s_r(u, v) : [0, 1]^2 \rightarrow \mathbb{R}$  denotes the *spatial loss field* (SLF) of emitter  $r$ , and  $c_{r,k}$  denotes the *power spectral density* (PSD) of emitter  $r$  at the  $k$ th frequency bin. Accordingly,  $\mathbf{c}_r \in \mathbb{R}^K$ , with  $[\mathbf{c}_r]_k = c_{r,k}$ , represents the discrete PSD of emitter  $r$ . Within the region, a set of sensors is deployed at spatial locations  $\Omega = \{(u_n, v_n) \in [0, 1]^2\}_{n=1}^N$ . Each sensor measures the received PSD vector  $\mathbf{f}(u_n, v_n) \in \mathbb{R}^K$ , whose  $k$ th entry satisfies  $[\mathbf{f}(u_n, v_n)]_k = f_k(u_n, v_n)$ .

The objective is to recover the collection of continuous functions  $\{f_k : [0, 1]^2 \rightarrow \mathbb{R}\}_{k=1}^K$  (or, in their compact form  $\mathbf{f} : [0, 1]^2 \rightarrow \mathbb{R}^K$ ) from the irregularly sampled measurements  $\{\mathbf{f}(u_n, v_n)\}_{n=1}^N$ , thereby reconstructing a spatially continuous radio map across all frequency bins. While the proposed framework can be readily extended to handle a continuous frequency domain, discretized frequency representations are standard in communication systems and are therefore adopted in this work. Fig. 1 illustrates the scenario.

### B. Low-rank and DGM Models for SC

Instead of treating  $f_k : [0, 1]^2 \rightarrow \mathbb{R}$  as a continuous function, [5], [8], [15] discretize the spatial domain and represent the radio map as a grid-based tensor  $\underline{\mathbf{X}} \in \mathbb{R}^{I \times J \times K}$  with

$$\underline{\mathbf{X}}(i, j, k) = f_k(u_i, v_j), \quad u_i = \frac{i-1}{I-1}, \quad v_j = \frac{j-1}{J-1}, \quad (2)$$

for  $i = 1, \dots, I$  and  $j = 1, \dots, J$ ; that is, each tensor grid corresponds to a specific spatial coordinate sampled uniformly

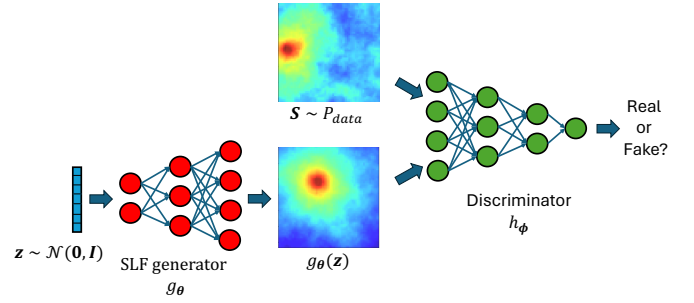


Fig. 2. GAN-based DGM training for SLF.  $\mathcal{D}_{\text{sim}}$  is a numerical simulator, e.g., [20]. After training, for each  $\mathbf{S}_r$ , there exists a  $\mathbf{z}_r$  such that  $\mathbf{S}_r = \mathbf{g}_\theta(\mathbf{z}_r)$  approximately holds; see more details in [5].

from the continuous region, including the boundary endpoints. (see the sensing grid in Fig. 1). Then,

$$\underline{\mathbf{X}} = \sum_{r=1}^R \mathbf{S}_r \circ \mathbf{c}_r \quad (3)$$

where the discrete SLF  $\mathbf{S}_r \in \mathbb{R}^{I \times J}$  and  $\mathbf{S}_r(i, j) = s_r(u_i, v_j)$ . The work [15] postulates that the discrete SLF  $\mathbf{S}_r$  has low rank and proposed a low-rank matrix completion approach. The work [8] uses a DGM to represent  $\mathbf{S}_r$ , i.e.,

$$\mathbf{S}_r = \mathbf{g}_\theta(\mathbf{z}_r), \quad (4)$$

where the DGM  $\mathbf{g}_\theta(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^{I \times J}$  is learned via auto-encoders [8] or GANs [5]; see Fig. 2 for an illustration of how a GAN-based DGM of SLFs is trained. Then, the formulation is

$$\min_{\{\mathbf{z}_r, \mathbf{c}_r\}} \sum_{(i,j) \in \hat{\Omega}} \left\| \underline{\mathbf{X}}(i, j, :) - \sum_{r=1}^R [\mathbf{g}_\theta(\mathbf{z}_r)]_{i,j} \mathbf{c}_r \right\|_2^2 \quad (5)$$

where  $\hat{\Omega}$  collects the on-grid measurement acquisition coordinates.

Compared to early kernel and interpolation based methods, e.g., [4], [6], [7], [11], the tensor based method in [15] provides recoverability under flexible, non-random sampling patterns. Compared to low-rank models, DGMs learn SLF's detailed characteristics from data and show much enhanced performance when the RF environment is complicated, e.g., with severe shadowing; see [5], [8].

The low-rank model in [15] was also combined with interpolation methods [10], [18] to handle off-grid sampling. The idea is to use off-grid samples to estimate  $\underline{\mathbf{X}}(i, j, k)$  first and then use (3) with  $\text{rank}(\mathbf{S}_r) \leq L$  to explain the estimated, on-grid  $\underline{\mathbf{X}}$ . Intuitively, the low-rank structures provide more restrictions to guide the interpolation (and vice versa). These approaches work reasonably to a certain extent, but the performance is limited by the accuracy of both interpolation and the low-rank model—both favor radio maps with less shadowing and smoother spatial characteristics.

## III. PROPOSED METHOD

We propose a DGM-based framework for off-grid SC. Unlike [9], [10], [18] that use low-rank models to interpret

interpolated on-grid points, we take an alternative: We assume the on-grid points are uniform samples of the continuous fields, and then convert the “uniform samples” to represent the entire field via DCT—whose approximation accuracy is well-understood for support-bounded signals. This way, we avoid using samples to estimate on-grid points. The approach also easily allows us to use pixel-based DGM in the pipeline.

#### A. Connecting On-grid SLFs to Off-grid Measurements

We begin by modeling the spatial loss field (SLF) of each emitter,  $s_r(u, v) : [0, 1] \times [0, 1] \rightarrow \mathbb{R}_+$ . We adopt a DCT-based representation for signals with bounded spatial support. As shown in [19], [21], non-periodic functions with compact support can be represented using Fourier series via appropriate even periodic extensions. Specifically, a univariate function  $f(x)$  supported on  $[0, 1]$  can be extended to an even, 2-periodic function, which admits a cosine-series expansion

$$f(x) = \sum_{k=0}^{\infty} \alpha_k \phi_k(x),$$

$$\phi_0(x) = 1, \phi_k(x) = \sqrt{2} \cos(k\pi x), k > 0,$$

where  $\{\phi_k\}$  form an orthonormal basis on  $[0, 1]$ . In practice,  $f(x)$  is approximated by a truncated series with cutoff  $K$ .

This construction extends naturally to two dimensions (or even higher dimensional tensor [19]). For  $s_r(u, v)$  supported on  $[0, 1]^2$ , applying even periodic extensions along each coordinate yields the tensor-product cosine representation

$$s_r(u, v) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} c_{i,j} \phi_i(u) \phi_j(v) \approx \sum_{i=0}^{I-1} \sum_{j=0}^{J-1} c_{i,j} \phi_i(u) \phi_j(v),$$

which is approximated in practice by truncating the series with cutoffs  $I - 1$  and  $J - 1$ . Under mild regularity conditions on the spatial loss field  $s_r(u, v)$ , the decay behavior of its DCT coefficients can be characterized using classical Fourier approximation theory [19], [21]. In particular, if  $s_r$  admits absolutely integrable mixed partial derivatives up to total order  $p$ , then its cosine-series coefficients  $c_{i,j}$  decay algebraically with index magnitude, i.e.,

$$|c_{i,j}| = \mathcal{O}((i^2 + j^2)^{-p/2}).$$

As a consequence,  $s_r(u, v)$  can be accurately approximated by retaining only low-index DCT coefficients, with the approximation error decaying polynomially in the truncation sizes.

**Remark 1** Note that functions admitting integrable mixed partial derivatives up to finite order  $p$  constitute a broad class of continuous functions. This class includes polynomial and smooth fields, as well as piecewise-smooth functions with localized irregularities or sharp transitions. As a result, the model remains expressive enough to capture realistic radio maps, which are typically governed by smooth path-loss variations combined with localized shadowing effects induced by obstacles.

We also define  $C \in \mathbb{R}^{I \times J}$  such that  $[C]_{i,j} = c_{i-1,j-1}$  for  $i = 1, \dots, I$  and  $j = 1, \dots, J$ . Or equivalently, in matrix form,

$$s_r(u, v) = \phi_I(u)^\top C \phi_J(v), \quad (6)$$

with  $\phi_D(u) = [\phi_0(u), \phi_1(u), \dots, \phi_{D-1}(u)]^\top \in \mathbb{R}^D$ . Let  $Q_u(i, :) = \phi_I(u_i)^\top$  for  $i \in [I]$  and  $Q_v(j, :) = \phi_J(v_j)^\top$  for  $j \in [J]$ , where  $u_i, v_j$  are defined in (2). Then, we have  $Q_u^\top Q_u = Q_u Q_u^\top = I_I$  and  $Q_v^\top Q_v = Q_v Q_v^\top = I_J$ .

#### B. DGM-based Off-grid SC: Our Formulation

As mentioned, most DGMs  $S_r = g_\theta(z_r)$  are trained using pixel-based models—that is, the training data and the representation are all on discrete grids. To connect  $S_r \in \mathbb{R}^{I \times J}$  with the continuous function  $s_r(u, v)$ , we use the orthogonality of  $Q_u$  and  $Q_v$ :

$$S_r = Q_u C Q_v^\top, C = Q_u^\top S_r Q_v. \quad (7)$$

Combining with (6), we have

$$s_r(u, v) = \phi_I(u)^\top Q_u^\top S_r Q_v \phi_J(v) \quad (8)$$

Given the off-grid measurements  $\{f(u_n, v_n)\}_{n=1}^N$  we can then formulate the recovery problem as follows

$$\min_{\{z_r, c_r\}} \sum_{(u_n, v_n) \in \Omega} \left\| f(u_n, v_n) - \sum_{r=1}^R s_r(u_n, v_n) c_r \right\|_2^2 \quad (9)$$

where  $s_r(u_n, v_n) = (\phi_I(u_n)^\top Q_u^\top g_\theta(z_r) Q_v \phi_J(v_n))$  and  $g_\theta(\cdot)$  is the pre-trained DGM as in Fig. 2. This formulation allows us to use off-the-shelf optimization algorithms, e.g., the alternating optimization approach in [8] to tackle the objective. Constraints on  $c_r$ , e.g., nonnegativity, can also be easily accommodated as in [5], [8].

**Remark 2** We note that the recent work in [17] uses a similar neural network-based formulation as in (9) but their  $s_r(u, v)$  is expressed by an implicit neural representation (INR):

$$s_r(u, v) = g_\beta(u, v), \quad (10)$$

where  $g_\beta : [0, 1] \times [0, 1] \rightarrow \mathbb{R}_+$  is an untrained neural network that maps continuous coordinates to a scalar value, and  $\beta$  is the optimization variable that should be learned together with  $c_r$  for  $r = 1, \dots, R$ . This approach essentially belongs to the untrained neural network learning category (see, e.g., [12], [22]), which is fundamentally different from (9). The former does not learn information from off-line data, but only use neural architecture to, hopefully, capture intricate data structure.

## IV. SIMULATIONS

**Data Generation.** We consider radio maps defined over a two-dimensional continuous spatial region covering  $63 \times 63$  m<sup>2</sup> and  $K = 64$  frequency bins. Our DGM is trained using SLFs with  $64 \times 64$  pixels, uniformly sampled from the spatial region. To simulate off-grid sensing, we acquire measurements over random locations from  $127 \times 127$  spatial grids. The high-resolution (HR) grids are also used for performance

TABLE I  
PSNR, SSIM, AND LNRE FOR DIFFERENT SAMPLING RATES (SRs).

| Method    | SR = 10%           |                    |                      | SR = 5%            |                    |                      |
|-----------|--------------------|--------------------|----------------------|--------------------|--------------------|----------------------|
|           | PSNR( $\uparrow$ ) | SSIM( $\uparrow$ ) | LNRE( $\downarrow$ ) | PSNR( $\uparrow$ ) | SSIM( $\uparrow$ ) | LNRE( $\downarrow$ ) |
| SCISP-LL1 | 29.87              | <b>0.8922</b>      | 0.0035               | 22.99              | 0.7610             | 0.0192               |
| GFRME     | 21.23              | 0.7631             | 0.0203               | 17.58              | 0.6886             | 0.0445               |
| Proposed  | <b>33.25</b>       | 0.8418             | <b>0.0024</b>        | <b>32.81</b>       | <b>0.8360</b>      | <b>0.0030</b>        |

evaluation—i.e., we check if the measurements and the  $64 \times 64$  DGM under (9) can recover the signal values over the HR grids. The SLFs are synthesized using the joint path-loss and spatial correlated log-normal shadowing model [20], while the PSDs are generated using a mixture of scaled Gaussian densities [8], [15]. The SLF generation parameters are set to a decorrelation distance of  $X_c = 100\text{m}$ , shadow variance  $\eta = 4\text{dB}$ , and path loss coefficient  $\alpha \sim \mathcal{U}(2, 2.5)$ . We use  $100K$   $64 \times 64$  SLFs for training the GAN generator; see [5] for the detailed architecture.

**Performance Metrics.** We employ three widely used metrics: peak signal-to-noise ratio (PSNR), structure similarity index (SSIM), and log-domain normalized reconstruction error (LNRE); see [5], [23], [24].

**Baselines.** We use the interpolation based tensor model in [18], namely, SCISP-LL1, and the INR based method in [17], GFRME, as our baselines. We use our method with a GAN-based DGM for SLFs as in [5]. We also use the standard Gaussian reweighting to  $C$  to suppress the Gibbs effect [25].

**Results.** We evaluate the proposed method on 100 radio maps with 6 emitters under sampling rates of 10% and 5%, and compare it with representative baselines.

Fig. 3 visualizes the ground truth and reconstructed radio maps over  $127 \times 127$  HR grids at randomly selected frequency bins. As shown in the figure, the proposed method consistently recovers the HR radio maps that closely match the ground truth in both global spatial structure and local details. In contrast, GFRME exhibits pronounced over-smoothing and loss of fine-scale spatial features, which is particularly evident under sparse sensing. Meanwhile, SCISP-LL1 introduces strong high-frequency artifacts and spatial oscillations, consistent with its reliance on interpolation and low-rank regularization.

Fig. 4 shows the ground truth and estimated  $\hat{f}(u, v)$  at two spatial locations, showing that the proposed method more accurately preserves the spectral shapes, while competing methods show visible estimation errors in spectral profile.

Table I reports performance metrics averaged over 100 test maps. Overall, the proposed method achieves the highest PSNR and LNRE among all competing approaches. While SCISP-LL1 attains slightly higher SSIM scores, its reconstructions exhibit noticeable spatial artifacts, as evident from the visual comparisons. Notably, the proposed method remains robust even at a 5% sampling rate, whereas the performance of competing methods degrades substantially.

## V. CONCLUSION

This work investigated radio map estimation from off-grid measurements over a continuous spatial domain. By leveraging a multi-dimensional DCT representation, we established

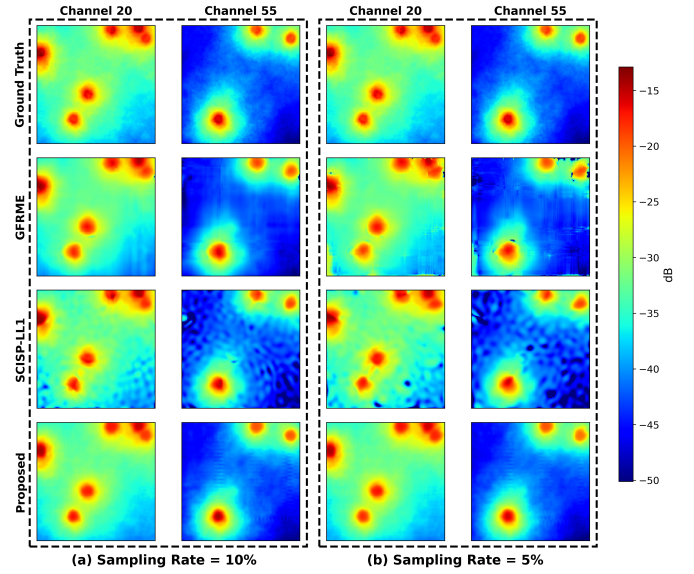


Fig. 3. Ground truth and reconstructed radio map by different methods at selected frequency bins under sampling rates of 10% and 5%.

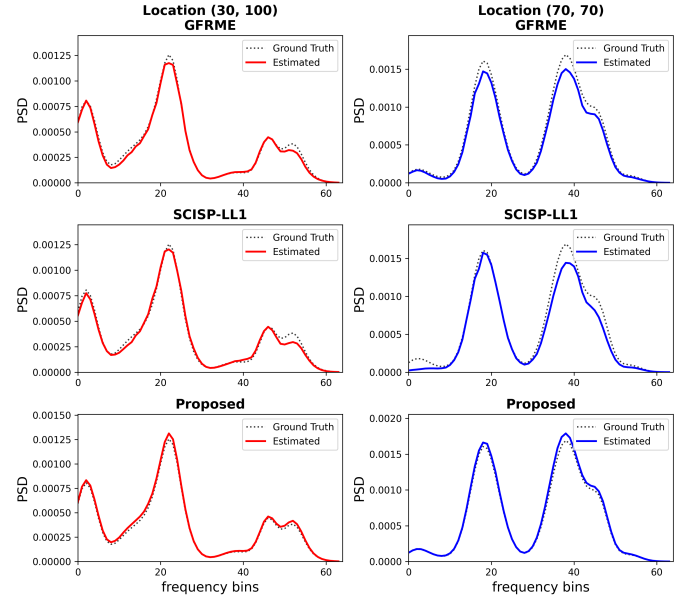


Fig. 4. Ground truth and estimated PSDs by different methods at two spatial locations under a 5% sampling rate.

a principled connection between off-grid sensing and deep generative models trained on gridded radio map data. The resulting formulation provides a compact and computationally efficient treatment of off-grid measurements, enabling off-the-shelf DGM-based spectrum cartography algorithms to be applied with simple modifications, without relying on spatial interpolation that can be unreliable in complex propagation environments. Numerical experiments demonstrated that the proposed approach achieves accurate reconstructions under sparse and irregular sampling, and consistently outperforms recent off-grid SC baselines.

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